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Explainable soft-voting classifier for heart disease prediction using SHAP and LIME

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Abstract

Heart disease remains a major global health challenge, underscoring the need for predictive models that are both accurate and interpretable to support early diagnosis and clinical decision-making. Using Shapley Additive Explanations (SHAP) and Local Interpretable Model-Agnostic Explanations (LIME) in this study allowed for a deeper level of understanding of model predictions as well as provided viable alternative ways to overcome the challenges associated with black-box classifiers. Using a soft-voting ensemble technique that combined representations for Random Forest (RF), Decision Tree (DT), and Support Vector Machines (SVM) was applied to evaluate a publicly available heart disease dataset, containing 5110 records with each record containing 76 attributes. A total of 14 clinically relevant features were identified and evaluated using an 80/20 train-test split for the experimental setup, with results further validated by 5-fold cross-validation. The proposed soft-voting ensemble model achieved an accuracy of 89.10%, precision of 83.90%, recall of 95.20%, and F1-score of 89.90%, out-performing all individual baseline models and many of the recent XAI-enhanced deep learning models. The combination of SHAP and LIME allowed for both global and local interpretations of the model's outputs, providing interpretable insights as to features like cp, thal, ca, and oldpeak contributed to the predictions and aligning these features with known cardiovascular risk factors. The results of this research demonstrate how interpretable (or explainable) ensemble-learning mechanisms can confer highly reliable, clinically useful heart disease risk predictions.

Keywords Explainable AI, Machine learning algorithms, SHAP, LIME, Clinical prediction models, Feature importance

1 Introduction

The healthcare industry plays a pivotal role in addressing the complex challenges associated with cardiac diseases such as heart failure, arrhythmias, cardiomyopathy, pericarditis, and many more. Among these, heart attacks remain one of the most common and life-threatening conditions, continuing to claim countless lives each year. From the study, we observed that cardiovascular diseases (CVDs) are universally acknowledged as the root cause of death, leading to the demise of around 17.9 million lives each year [1]. It is noted that more than four-fifths of CVD deaths result from heart attacks, with a

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significant part occurring prematurely in individuals below the age of 70 [2]. On average, one person loses their life to CVD every 34 s [3].

Timely diagnosis, effective treatment, and comprehensive preventive measures are essential components of healthcare strategies aimed at combating heart disease. Healthcare professionals, including cardiologists, nurses, and primary care physicians, work collaboratively to assess risk factors, perform diagnostic tests, implement evidence-based treatments, and provide ongoing care and support to individuals affected by heart disease. Beyond clinical care, healthcare systems must prioritize public health initiatives focused on promoting heart-healthy lifestyles, raising awareness of risk factors such as smoking, obesity, and unhealthy lifestyles, and ensuring access to affordable and equitable healthcare services for all individuals. A comprehensive approach, which includes prevention, early detection, and innovative treatments, plays an important role in reducing heart disease problems and improving health outcomes globally.

In recent years, advancements in technology, particularly Artificial Intelligence (AI), have opened new possibilities for enhancing cardiovascular care. The rise of autonomous machines includes technological advances that have been made through automation and intelligence navigation systems; thus, AI innovations will revolutionize the world across many industries such as agriculture where autonomous farming machines are being developed through intelligent navigation technology as well as robotics [4]. AI empowers healthcare professionals to use predictive modeling, personalized treatment recommendations, and efficient patient monitoring. Even though great strides have been made in the development of heart disease prediction models based on machine learning (ML) techniques, the majority of these models remain "black boxes." Because of limited interpretability of existing models, there is still significant reluctance to adopt these technologies within the clinical community. An important gap exists between these highly accurate models and the creation of new models that not only provide high degrees of accuracy but also provide users with an explanation of why the model arrived at its prediction.

To address these challenges, Explainable AI (XAI) has emerged as a critical complement to traditional AI methods [5, 6]. XAI provides transparent, human-understandable explanations for model outputs, making it easier for clinicians to interpret and trust the results. While previous studies employ SHAP and LIME independently to interpret ML models, few efforts have systematically integrated an ensemble approach using multiple XAI methods for better predictions of heart disease in structured data. [4]. The integration of AI and XAI in healthcare not only enhances diagnostic accuracy and treatment efficacy but also supports proactive preventive measures. AI-driven predictive analytics can identify individuals at high risk of developing heart disease by analyzing the genetics, lifestyle, and clinical factors, enabling targeted interventions to mitigate risk factors and prevent disease progression [7]. By leveraging the capabilities of AI and XAI in the fight against heart disease, healthcare systems can revolutionize the delivery of cardiovascular care, making it more precise, proactive, and patient-centered.

After understanding the severity of the above-mentioned problem, we have employed ML algorithms, including Random Forest (RF), Decision Tree (DT), and Support Vector Machine (SVM) to address the challenges posed by CVD. In addition, we have used the XAI techniques such as SHAP and LIME to gain insight into the predictions made by ML algorithms. This research study's primary focus is on establishing an ensemble

soft-voting classifier that can improve predictive accuracy through the combination of additional tools that provide insight into how predictions are made. This improved performance will be evidenced through the greater ability to interpret the clinical rationale behind the predictions of these models than when using the baseline individual model.

This comprehensive approach aims to improve the transparency, interpretability and effectiveness of predictive models for early detection, intervention, and management of CVD, thus contributing to improved healthcare outcomes and reduced.

1.1 Contributions

The contributions of this paper are described as follows.

1. An ensemble soft voting classifier including RF, DT, and SVM is utilized for the detection of heart disease.
2. The SHAP and LIME techniques are used, which employ an XAI to improve the ability of medical professionals for making decisions and more accurate assessments.
3. The proposed model is assessed using the standard heart disease dataset. Simulation results show that the proposed technique has demonstrated promising outcomes as compared to traditional baseline models.

1.2 Organization

The structure of the paper is as follows. Section 2 summarizes previous work and identifies the specific issues that need to be addressed. Section 3 details the proposed approach, followed by the performance evaluation and experimental results. Section 4 discusses the findings and their implications. Finally, Sect. 5 concludes the paper and provides directions for future research.

2 Related work

In this section, we have discussed the previous research on the application of AI and XAI in healthcare systems. By incorporating attention processes into convolutional neural networks (CNN), current developments in deep learning have shown significant gains in feature extraction. As demonstrated by models like DeepLabv3Att in road-segmentation tasks, research indicates that integrating attention modules with CNN architectures can greatly improve representational capacity [8]. For instance, Wani et al. in [9] proposed the DeepXplainer model, which combines CNN and XGboost for lung classification and implemented SHAP for evaluating the feature importance in the entire dataset. The authors have utilized the "Survey Lung Cancer" open-source dataset, and the DeepXplainer model was applied and yielded superior performance compared to existing methods across multiple metrics such as accuracy, sensitivity and the F1-score. Following the model's highly accurate predictions, each prediction is interpreted through the implementation of XAI techniques at all levels. Similarly, the study mentioned in [10] introduces a novel hybrid deep learning (DL) framework that combines a convolutional neural network (CNN) and long short-term memory (LSTM) for the purpose of detecting CVD. Their evaluation results on a publicly available dataset demonstrate a high accuracy of 73.52% with feature engineering and 74.15% without feature engineering. Combining conventional machine learning classifiers like SVMs with convolutional network-based feature extraction techniques has also shown success when attempting to identify various types of diseases for the purpose of providing diagnostic assistance. An

example of this research includes enhanced MRI images of alzheimer's disease patients [11]. Additionally, the model in the research incorporates XAI techniques *i.e.,* SHAP to identify the key features associated with CVD, offering insights into the important factors influencing disease prediction.

In addition, the research in [12] undertakes a comparative examination of predictive features extracted from various classification algorithms, namely AdaBoost Classifier (ABC), RF, Gradient Boosting Classifier (GBC), and Light Gradient-Boosting Machine (LGBM), to uncover shared patterns in predictive results on heart attack dataset. Their results show that the LGBM stands out as the top performer among these classification algorithms, achieving an impressive average training accuracy of 99.33%. Additionally, a stacking-based ensemble framework, in combination with tree-selection techniques, can also yield highly predictive results while maintaining model explainability in the diagnosis of liver diseases [13]. Anand et al. [14] proposed the ST-CNN-GAP5-Spatio Temporal- Convolutional Neural Network-Global Average Pooling model, which surpassed the existing state-of-the-art results on the PTB-XL dataset, attaining an AUC of 93.41%. With this, the authors in [15] proposed a novel automated classification algorithm that integrates multiple deep-learning approaches for the detection of prostate cancer using ultrasound (US) and magnetic resonance imaging (MRI) modalities, achieving 97% and 80% accuracy, respectively. The model incorporates XAI to explain its benign or malignant classifications. Moreover, advanced XAI techniques combined with ensemble-based neural network models have shown clear diagnostic abilities in identifying liver diseases with improvements regarding accuracy and interpretability [16].In addition to traditional ensemble learning techniques, hybrid stacking models that utilize both CNN-based feature extraction in combination with boosting models have performed very well on c0omplex classification tasks including plant leaf disease detection, as demonstrated by Hossain et al. [17].

The authors in [18] developed a Python-based application, focusing on handling categorical variables and converting categorical columns. In this, an RF classifier algorithm was developed to enhance the accuracy of identifying heart diseases, achieving an approximate accuracy rate of 83% over training data. Similarly, in [19], the authors introduced a highly accurate and interpretable ML model employing an RF classifier in detecting Alzheimer's disease. In this paper [20], the authors introduced an automated tuberculosis (TB) detection system leveraging the latest models of deep learning (DL) that employs advanced segmentation networks to identify multimedia CXR zones of interest.

The authors in [21] presented a ML framework designed to accurately classify DaTSCAN images as indicative of Parkinson's disease or not. To achieve this, they have employed visual indicators generated through the Local Interpretable Model-Agnostic Explainer (LIME) technique. Interpretable ML allows for a clear understanding of the importance of specific features during the Hepatitis C testing process because of the combination of using deep feature synthesis and integral ML algorithms [22]. With this, the authors in [23] introduced an XAI technique to employ ML approaches for analyzing COVID-19 metagenomic next-generation sequencing (mNGS) samples. Similarly,in the research [24], the authors utilized AI to diagnose COVID-19 mass spectrometry (MS) data, integrating XAI for decision-making at both individual and collective levels. In this research [25], ML methods were employed for Parkinson's disease (PD) diagnosis

by incorporating gene expression data sourced from diverse origins and applying XAI techniques to identify crucial pieces of gene features influencing disease discovery. The authors in [26] investigated various ML classification methods, including DT, SVM, RF, Logistic Regression, KNN, and ensemble techniques, the proposed system achieved optimal performance using the XGBoost classifier with the ADASYN [27] for class imbalance approach, attaining 81% accuracy, F1 coefficient value of 0.81and AUC of 0.84.

Recent research has shown that ML integration with XAI techniques for clinical decisions is effective. For instance, combining ML with SHAP and LIME produces more interpretable dengue classification results compared to using either method alone [28] and using ensemble approaches to XAI significantly enhances maternal health risk prediction [29]. These results validate the significance of explainable ensemble architecture within the modern healthcare analytical environment. From the literature survey, we have observed that the existing ML models perform accurately, but they often operate as "black boxes", which makes it difficult to understand decision-making. Thus, these models lack transparency, which will limit their use in clinical practice. Therefore, the proposed model combines the strengths of a deep convolutional neural network (DCNN) with a gated recurrent unit (GRU), while integrating XAI techniques. Specifically, the DCNN-GRU model harnesses the capabilities of CNNs to automatically extract pertinent features from lung images, thereby capturing both local and global patterns. These extracted features are then input into a GRU, which adeptly models temporal dependencies and encapsulates sequential information inherent in lung images. Moreover, the incorporation of XAI techniques such as LIME, SHAP, and Grad-CAM strengthens the interpretability and transparency of the model.

3 System model

Figure 1 shows a 4-layer system model designed for the diagnosis of heart disease with XAI. Each layer focuses on different aspects like interaction, functionality, and transparency. The 4 layers- recipient layer, clinical layer, diagnosis layer and transparency layer- are described in the following sub-sections.

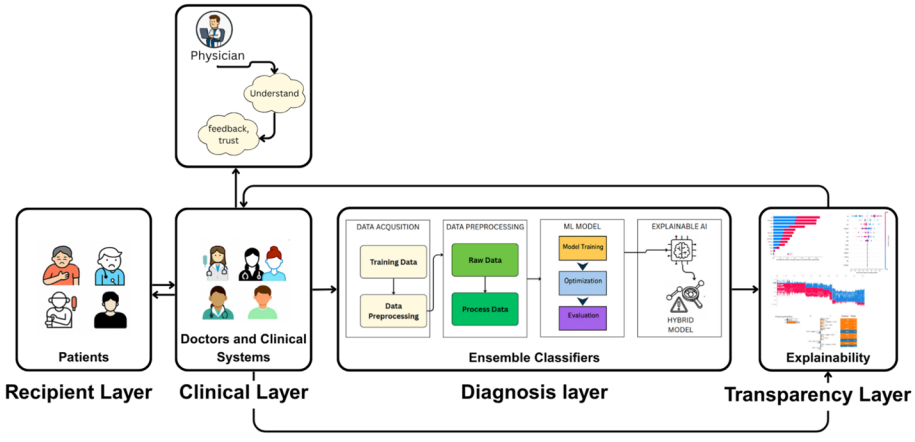


Fig. 1 Proposed system model

3.1 Recipient layer

The recipient layer in the proposed system model signifies the layer that particularly handles and manages the diverse patient population, represented by the variable 'n.' The patient population represented by 'n' has a wide range of ages, various sex, and health histories. These diverse parameters collected by competent healthcare professionals offer a broad understanding of how various factors can impact health outcomes and care needs. This layer acts as the interface between the healthcare system and patients, focusing on collecting data, analyzing, and addressing the health needs and behaviors of this heterogeneous group. These clinical sessions are beneficial for allowing thorough health assessments, giving healthcare professionals the opportunity to complete in-depth evaluations. Healthcare professionals leverage these interactions to reveal valuable diagnostic insights, helping them identify underlying medical conditions in patients with complex health profiles.

3.2 Clinical layer

In this layer, a collaborative team of medical professionals, including a large number of physicians, plays a crucial role. This team is supported by a dedicated group of healthcare providers such as nurses, technicians, advanced computer systems, and other necessary support staff. This teamwork is necessary for the efficient operation of the health facility. Once a patient acquires an appointment with a medical professional, a thorough diagnosis process is initiated, aimed at supporting informed decision making. The patient receives a detailed examination that involves physical examinations, vital sign checks, and an analysis of the patient's symptoms. The data collected during this assessment are meticulously recorded and stored in a database, creating a detailed and comprehensive record of the patient's health condition.These data are stored for ongoing analysis, supporting subsequent decisions regarding medical procedures, treatments, and diagnostic tests. The detailed documentation and archival practices followed in this phase ensure that future healthcare interventions are based on a clear and well-recorded understanding of the patient's health history.

3.3 Diagnosis layer

The diagnosis layer refers to a stage within the system model that focuses on interpreting and analyzing the collected data to determine a patient's health condition. In this, a classification process is carried out using distinguished, pre-trained model designed to detect complex patterns in the input data. Once the patient's information is entered into the model, it undergoes a set period of analysis during which the model performs intricate calculations to generate a conclusive result. If the prediction of the model matches the CVD of the patient, a series of coordinated actions is triggered. Conversely, if the prediction of the model does not suggest CVD, an alternative critical path is followed. This explanatory process helps healthcare professionals and other stakeholders understand the rationale behind the decision of the model.

3.4 Transparency layer

After the diagnostic layer identifies whether a patient has CVD, the next step is to transfer the model's results, along with the patient's key health indicators and symptoms, to the explainability layer. This layer is responsible for providing clear explanations of the

model's outcomes. The goal is to clarify the reasoning behind two possible outcomes: the detection of CVD or its absence. The healthcare provider then communicates the findings to the patient with clarity and empathy, ensuring that the reason behind the diagnosis is well understood. It helps to foster trust and understanding between the patient and the healthcare team.

3.5 Computational pipeline

The entire computational workflow employed in this research is described in this section. In Sect. 4.1, the preprocessing approach involved removing duplicates, handling missing values, one-hot encoding of categorical variables, and feature scaling when the data was obtained is explained. Only 14 clinically significant features were finally chosen from the original dataset's 76 variables in order to create the final models. The data sets were subsequently used to train the three base classifiers, namely: SVM, DT and RF—the details of which are shown in Sect. 4.2. The probabilistic outputs from each classifier were incorporated using a soft voting ensemble method described in Sect. 4.2.4. The ultimate predictions were derived from interpretations using SHAP and LIME as detailed in Sect. 4.3. This computational pipeline allows users to see the entire process from raw data acquisition through to the final output of interpretability and predictions.

4 Proposed method

4.1 Data acquisition and pre-processing the dataset

In this research, the Heart Disease dataset [30] is utilized. This dataset comprises records dating from 1988, encompassing four famous databases: Cleveland, Hungary, Switzerland and Long Beach V. It has 5110 rows with 76 attributes which has crucial information for heart disease prediction. The predominant focus of the work in this research is on a subset of 14 attributes, in order to pinpoint the most informative features for a precise diagnosis. The "target" field is indicative of the existence of heart disease in the patient. 'No disease' is represented by integral value '0' and 'presence of disease' is represented by value '1'.

The significant part of the analysis and ML pipeline is the preprocessing of the data. The analysis and model training involve cleaning and transforming raw data into a suitable format. This process includes eliminating duplicates, managing missing values, encoding categorical variables, and scaling features. After importing the required libraries, a one-hot encoding technique is implemented to preprocess data to represent categorical variables as binary vectors. Since there was no significant imbalanced class in the training dataset, there was no need to consider any additional resampling methods. Validation of our model was accomplished through 5-Fold Cross Validation processes. By addressing these issues, data preprocessing enhances the quality and reliability of the dataset, resulting in more precise and robust results in subsequent analytical and ML tasks. Proper data pre-processing contributes significantly to the overall success of data-driven projects by ensuring that input data is prepared in a way that increases the effectiveness of algorithms and models. The description of the features in the dataset is given in Table 1 .

Table 1 Details of the features in the heart disease dataset

Feature name	Full name	Description
Age	Age of the patient in years	Basic demographic variable representing the age of the patient
Sex	Gender of the patient	Demographic variable representing 1 as male and 0 as female
Cp	Chest pain	Categorizes the type of chest pain as 0: Typical angina, 1: Atypical angina, 2: Non-anginal pain, 3: Asymptomatic
Trestbps	Resting blood pressure	Blood pressure measured when the patient is at rest (mm Hg)
Chol	Serum cholesterol	Represents the total cholesterol level in the blood
Fbs	Fasting blood sugar	Indicates whether fasting blood sugar level is > 120 mg/dL
Restecg	Resting electrocardiographic results	Results of an ECG/EKG taken while the patient is at rest
Exang	Exercise-induced angina	Indicates whether angina occurred as a result of physical exertion
Oldpeak	ST depression induced by exercise relative to rest	Used in stress testing to evaluate presence of ischemia
Slope	Slope of peak exercise ST segment	Pattern of the ST segment during peak exercise, used in diagnosis
Ca	Coronary artery count	Indicates the extent of coronary artery disease
Thal	Thalassemia	Blood disorder with lower-than-normal hemoglobin and RBC count
Target	Outcome variable	Indicates the presence or absence of heart disease

4.2 Model architecture

The proposed methodology implements an ensemble based approach intelligently leveraging multiple ML algorithms, including SVM, DT, and RF classifiers. These algorithms are combined using a soft voting classifier to improve accuracy. The explanation of each algorithm is provided in the following subsections.

4.2.1 SVM

SVM is a ML technique utilized for supervised learning to solve complex classification and regression tasks [31]. Its essential objective is to find an optimal hyperplane that servers as a decision boundary to segregate the space into categories where data points can be classified into distinct classes. The hyperplane is strategically chosen to maximize the margin, which is the distance between the hyperplane and the nearest data points from each class. These vectors play a crucial role in defining the orientation and position of the hyperplane, enabling the classification boundary to maximize the margin. By relying on these critical data points, SVM simplifies computations and improves its capacity to generalize to new, unseen data. The function of hyperplane line is given by Eq. 1:

$$ax + by + c = 0 \tag{1}$$

Renaming the variable x as x_1 and y as x_2, the equation becomes:

$$ax_1 + bx_2 + c = 0 \tag{2}$$

If $\mathrm{x} = (x_1, x_2)$ is defined as the input vector and $\mathrm{w} = (a, b)$ as the weight vector, the equation can be written in vector form as given in Eq. 3:

$$\mathrm{w} \cdot \mathrm{x} + c = 0 \tag{3}$$

Here, $\mathrm{w} \cdot \mathrm{x}$ represents the dot product of the vectors w and x, and c is a bias term (equivalent to $-b$ in some representations).

Using the Eq. 3, the hypothesis function $h(\mathrm{x}_i)$ for classification is defined as in Eq. 4:

$$h(\mathrm{x}_i) = \begin{cases} +1 & \text{if } \mathrm{w} \cdot \mathrm{x}_i + c \geq 0 \\ -1 & \text{if } \mathrm{w} \cdot \mathrm{x}_i + c < 0 \end{cases} \tag{4}$$

4.2.2 DT

It is a supervised ML technique, used for both classification and regression tasks [32]. It is structured as a tree, where the internal nodes represent properties or attributes of the dataset, the branches correspond to decision rules, and the leaf nodes represent the outcomes. Decision trees can handle both numerical and categorical data.

Attribute Selection in DTs: Randomly selecting the root node or internal nodes from the dataset can result in an inaccurate tree structure. Therefore, specific criteria are followed for attribute selection to ensure optimal tree formation:

a. *Entropy:* Entropy measures the randomness or uncertainty in the dataset [32]. A higher entropy value indicates greater randomness, making it harder to gain meaningful information. It is mathematically defined in Eq. 5:

$$\text{Entropy}(S) = -\sum_{c \in C} p(c) \log_2 p(c) - \sum_{d \in D} q(d) \log_2 q(d) \tag{5}$$

where:

- S: Dataset on which entropy is calculated.
- C: Set of classes in S.
- $p(c)$: Proportion of data points associated with class c relative to the total number of data points in S.

b. *Information Gain:* Information gain is a statistical metric that evaluates how well a given attribute separates the training examples with respect to the target classification [33]. It is calculated as Eq. 6:

$$\text{Information Gain}(S, a) = \text{Entropy}(S) - \sum_{v \in V} \frac{|S_v|}{|S|} \text{Entropy}(S_v) \tag{6}$$

where:

- a: Specific attribute or class label.
- S_v: Subset of S corresponding to the attribute value v.
- $\frac{|S_v|}{|S|}$: Proportion of values in S_v relative to the total number of values in S.
- $\text{Entropy}(S_v)$: Entropy of the subset S_v.

A well-formed decision tree is characterized by a high information gain and a low entropy value, ensuring effective classification and regression.

4.2.3 RF

RF is designed to address complex classification and regression problems by combining multiple classifiers [34]. It consists of numerous DTs that share the same structure of nodes but are trained on different subsets of data, resulting in varied outputs at the leaf nodes. This diversity among trees enhances the overall performance of the model.

Unlike a single decision tree, RF aggregates predictions from all the individual trees and makes the final decision based on majority voting. The accuracy of the model improves with an increasing number of decision trees, as it reduces the risk of overfitting and provides more robust predictions.

When using RF for data classification, the Gini index [35] is often utilized to measure the impurity of nodes. The Gini index is defined as in Eq. 7:

$$\text{Gini} = 1 - \sum_{i=1}^{C} (p_i)^2 \tag{7}$$

where:

- p_i: The relative frequency of observing class i.
- C: The total number of classes in the dataset.

This formula calculates the Gini index for each branch of a node, providing a measure of how well the node splits the data into distinct classes. A lower Gini index indicates a better split, contributing to more accurate predictions.

4.2.4 Proposed soft voting ensemble approach

A voting classifier is a metaclassifier that combines similar or diverse ML models to make predictions based on a majority voting mechanism [36]. The voting classifier has two distinct approaches: hard voting and soft voting. In hard voting, the final prediction is made by selecting the class that appears most frequently among the base models, based on a majority vote. In contrast, soft voting relies on the predict_proba method of the base models, where the predicted probabilities are averaged to determine the final output. By intelligently combining the predictions from different models, voting classifiers aim to enhance the overall performance, often outperforming individual base models. In this work, the soft voting classifier is used. The steps involved in the soft voting classifier are outlined below.

4.3 Training data preprocessing

The training dataset $D = \{(x_i, y_i)\}_{i=1}^{n}$ is shuffled to ensure unbiased training of individual classifiers. Each data point x_i is passed to the base classifiers SVM, DT, and RF. SVM is used because it is capable of capturing complex decision boundaries, which increases performance, and DT helps reduce variance and improve accuracy when both are incorporated into an ensemble method. RF is chosen because it helps maintains low bias, and performs effectively with high-dimensional data.

4.4 Probability estimation

Each model independently computes the probability $P_i(c \mid x)$ for each target class c using the `predict_proba` method [37]:

$$P_i(c \mid x) = \text{Probability of class } c \text{ by model } M_i.$$

where $P_i(c \mid x)$ represents the probability of class c predicted by the i-th model.

4.5 Aggregation of probabilities

The final probability for each class c is computed as a weighted average of the individual probabilities given in Eq. 8:

$$P_{\text{final}}(c \mid x) = \sum_{i=1}^{3} w_i \cdot P_i(c \mid x) \tag{8}$$

where:

- w_i: Weight assigned to the i-th model, reflecting its reliability or performance ($w_i \geq 0, \sum_{i=1}^{3} w_i = 1$).
- $P_i(c \mid x)$: Probability predicted by the i-th model for class c [38].

Equal weights ($w_1 = w_2 = w_3 = \frac{1}{3}$) were assigned to SVM, DT, and RF. This is due to the fact that there was very little difference in favour of one classifier over another during training and cross-validation, thus making it appropriate and stable for the dataset, as all three base classifiers displayed similar levels of predictive behaviour throughout training and cross-validation.

4.6 Final prediction

The class with the highest aggregated probability is selected as the final prediction as given in Eq. 9:

$$\hat{y} = \arg\max_{c} P_{\text{final}}(c \mid x) \tag{9}$$

The proposed ensemble approach is shown in the Fig. 2. This figure shows the design of the proposed ensemble model, which combines the SVM, DT, and RF classifiers using an intelligent soft voting methodology. Each classifier would produce probabilistic predictions that, in turn, would be averaged together to create a final output. This process combines classifier predictions and provides improvements through accuracy and robustness. The diagram gives a clear depiction of the flow of the ensemble model; from inputting the data through the individual models' processing, and to the final decision on output. It demonstrates the modularity and interpretability of the pipeline for the intended audience.

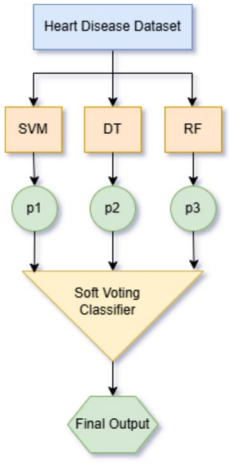


Fig. 2 Proposed soft voting ensemble approach

Table 2 Comparison performance of proposed and baseline ML models

Model	Acc. (%)	Precision (%)	Recall (%)	F1 Score (%)
SVM	83.90	80.33	91.58	85.59
DT	86.34	80.62	97.20	88.14
RF	85.37	83.48	89.72	86.45
Ensemble	89.10	83.90	95.20	89.90

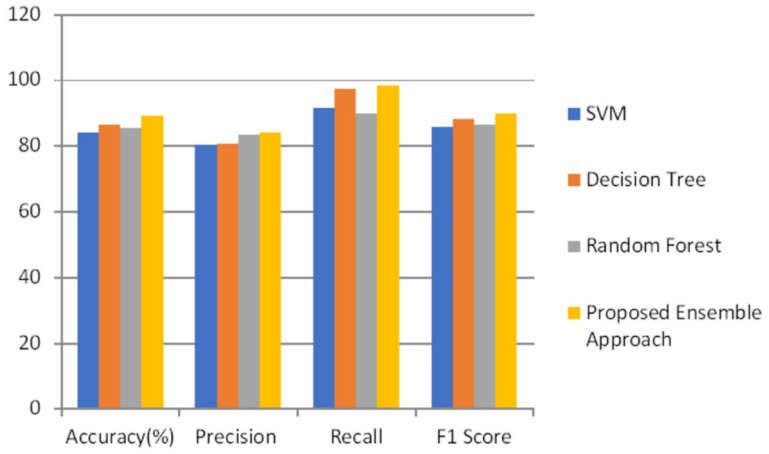


Fig. 3 Comparison of performance evaluation of proposed and baseline ML models

The experiments were carried out on the heart disease dataset. The dataset was split into training and testing sets, with 80% allocated for training and 20% for testing. To evaluate the performance and reliability of the proposed approach, metrics such as precision, precision, recall, and F1 score were employed which are calculated using the following formulas given in Eqs. 10, 11 and 12:

$$\text{Precision} = \frac{\text{TP}}{\text{TP} + \text{FP}} \tag{10}$$

$$\text{Recall} = \frac{\text{TP}}{\text{TP} + \text{FN}} \tag{11}$$

$$F1 = 2 \cdot \frac{\text{Precision} \cdot \text{Recall}}{\text{Precision} + \text{Recall}} \tag{12}$$

Table 2 shows a comparative analysis of the baseline ML models and the proposed soft voting ensemble approach. From Table 2, the proposed soft voting ensemble method has achieved maximum accuracy, precision, recall, and F1 score values of 89. 10%, 83. 90%, 95. 20% and 89. 9%, respectively. It can be seen from the table that SVM and RF performed comparatively less effectively on the data set with an accuracy value of 83. 90% and 85. 37%, respectively. The soft voting ensemble method has achieved the maximum accuracy of 89.10%. The comparison of baseline performance evaluation with the proposed approach is shown graphically in Fig. 3.

In Fig. 3, bar graph illustrates the overall performance metrics—accuracy, precision, recall, and F1 score—across single models (SVM, DT, RF) and then the ensemble classifier. The ensemble ultimately performed better than the single models from all statistics, but especially in accuracy for overall performance. The figure provides a clear

preliminary quantitative comparison that highlights the superior predictive performance of the ensemble approach.

4.7 XAI

XAI facilitates the interpretation of complex ML models and provides insight into patterns and features that contribute to the prediction outcomes [39]. In the context of heart disease prediction, XAI enables healthcare professionals to comprehend the factors that influence the risk assessment of a patient, helping to make informed clinical decisions. By explaining the reasoning behind the predictions, XAI not only enhances the trust and acceptance of AI systems in healthcare, but also empowers clinicians to validate and refine predictive models based on domain knowledge and patient-specific factors. In addition, transparent AI models promote patient participation by facilitating communication and comprehension of predictive results, thereby promoting shared decision-making between healthcare providers and patients. Thus, the adoption of XAI in the prediction of heart disease not only improves the accuracy and reliability of predictive models, but also improves their usability and acceptance in clinical practice, ultimately contributing to better patient outcomes and healthcare delivery. In this research, we employed SHAP and LIME models as XAI techniques to interpret results [40, 41]. These two models are explained, and the outcomes are presented in the subsequent section.

4.7.1 SHAP

SHAP is a familiar XAI technique that explains the outcomes of the ML model [42]. This approach enables recognition of how the model generates a particular prediction and identifies the features that play the most dominant role in determining the forecast. Therefore, SHAP can be said to be used mainly to detect and correct biases and improve performance. It is achieved by computing the SHAP values to see the insightful features, and these insights form the basis for further analysis and corrective measures. The SHAP value can be determined using Eq. 13:

$$\varphi_i(f,x) = \sum_{z' \subseteq x'} \frac{|z'|!(M-|z'|-1)!}{M!} \left[f_x(z') - f_x\left(\frac{z'}{i}\right) \right] \tag{13}$$

where,

$$\begin{aligned}
\varphi_i &= \text{Shapley value for feature } i \\
f &= \text{Blackbox model} \\
x &= \text{Input data point} \\
z' &= \text{Subset} \\
x' &= \text{Simplified data input} \\
f_x(z') &= \text{Model output with feature } i \\
f_x(\frac{z'}{i}) &= \text{Model output without feature } i
\end{aligned}$$

It evaluates the contribution of a single attribute by considering all potential subsets of features in which it is included. The contribution is then weighted by the number of subgroups with the feature and averaged over all capable subsets. A single SHAP value

is produced for individual features to explain the model prediction for a particular instance.

4.7.2 LIME

LIME focuses on explaining the prediction of the model for individual instances, and its primary goal is to provide human-understandable and interpretable descriptions of the predictions made by complex machine learning techniques [43]. This XAI model not only visualizes the prediction, but also gives an explanation of the individual prediction by focusing on the local area of the particular prediction. Furthermore, LIME can easily produce clear explanation which is logical in the particular local region. Hence, the rest of the model is not interfered with and still gets a valid explanation of why a particular prediction is made for an instance [44]. Any black box model can be explained by creating such a local approximation. To further explain individual predictions, LIME targets training local surrogate techniques rather than training a global surrogate model. Mathematically, local surrogate models with interpretability constraints can be depicted in Eq. 14

$$\text{explanation}(x) = \arg\min_{g \in G} \mathcal{L}(f, g, \pi_x) + \Omega(g) \tag{14}$$

Here, x is the chosen ML model, g is the parameter that minimizes loss $\mathcal{L}$ which measures how close the explanation is to the prediction of the original model f, while the model complexity $\Omega(g)$ is kept low. G is the family of possible explanations. The proximity measure π_x defines how large the neighborhood around instance x which is considered for the explanation. In practice, LIME only optimizes the loss part. The following algorithm shows the implementation of LIME.

Algorithm

1. Classifier f, Number of samples N
2. Instances x and its interpretable version x'
3. Similarity kernel π_x, length of explanation $KZ \leftarrow \{\}$

for $i \in \{1, 2, 3, \ldots, N\}$ **do**
$z_i' \leftarrow$ sample around (x')
$Z \leftarrow Z \cup \langle z_i', f(z_i), \pi_x(z_i) \rangle$ **end for**
$\omega \leftarrow K - \text{Lasso}(Z, K)$ with z_i' as feature, $f(z)$ as target
return ω

4.8 Results and discussion

This section discusses the findings related to interpretability experiments on the dataset. The findings can be categorized into two areas: local and global. The subsections that follow will expand on each of the aforementioned categories and highlight the practical and methodological limitations of explainability and model performance.

4.8.1 Local findings

The Fig. 4 illustrates a local explanation for a single prediction instance using SHAP [45]. It visually demonstrates how individual features impacted the final classification outcome for a specific patient—whether they are diagnosed with heart disease or not. Note

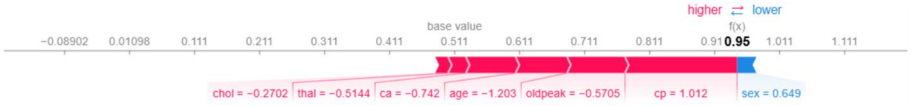


Fig. 4 Local instance wise plot of the proposed ensemble approach

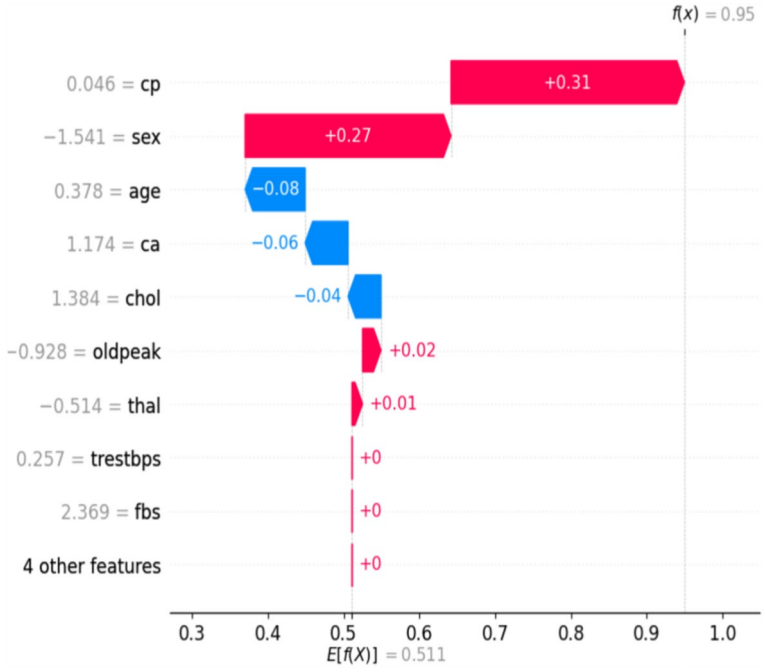


Fig. 5 Waterfall plot of the proposed ensemble approach

that the plot uses directed bars to communicate positive (increasing risk) and negative (decreasing risk) contributions from each individual input feature. For example, a high level of 'cholesterol' and / or an abnormal value of 'oldpeak' (ST depression induced by exercise) may have moved the model toward predicting disease, while the normal 'restecg' and the absence of 'exercise-induced angina' may have reduced the risk. Local interpretability is vitally important in the clinic because this is what doctors do when they weigh patient characteristics in front of them for diagnosis. The local transparency provided by the model enables healthcare practitioners to establish trust with the model and validate the AI's decision based on medical rationale.

Figure 5 shows the SHAP waterfall plot that provides additional analysis of local prediction by showing the cumulative contribution of each feature to the model output for a specific patient. It starts from the model's base value or average prediction from the data set and adds contributions of each specified feature to arrive at the final output probability [46]. The features that positively contribute to the prediction (indicating a higher risk of heart disease) are represented in red, while the negatively contributing features are represented in blue. The waterfall plot outlines a clear journey from a neutral prediction to how the outcome (final output probability) is arrived at, similar to transparent reflective diagnostic reasoning from a clinician. For instance, a high "ca" (number of major vessels colored by fluoroscopy) may add a significant amount to the prediction of heart disease, while a low "cholesterol" may subtract a significant amount. Since the waterfall shows a stepwise path, this also allows clinicians to clearly see how the model has come

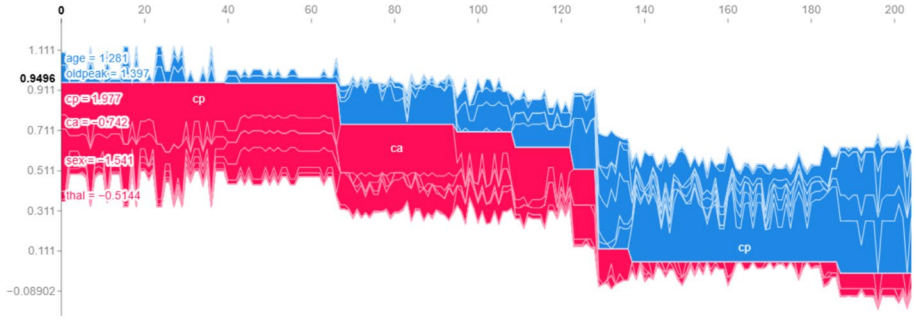


Fig. 6 Global (Summary) ensemble approach

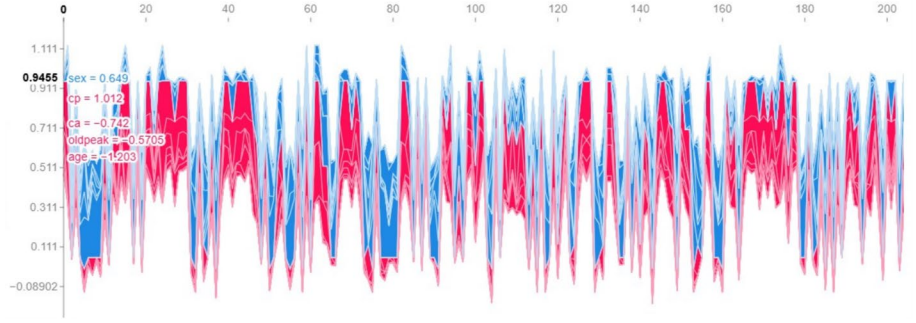


Fig. 7 Global (Ordering) plot of the proposed ensemble approach

to its conclusion and use this to give a clearer explanation of the outcome for patients. The increased credibility and trust in AI makes the use of AI more ethically responsible.

4.8.2 Global findings

The Fig. 6 depicts the SHAP global summary plot and provides a high-level visualization of the influence each feature has on the model predictions, throughout the data set [47]. It ranks each feature by its average influence on the model output (SHAP value), while color-coded scatter points show how the specific features were related (in terms of low-high) to shifts in prediction. For example, high (red) values of "oldpeak," or "cp" (chest pain type) might dominate for instances that became classified as high-risk, while low (blue) values for "ca" might correlate to low-risk classifications. This plot is essential for identifying the features that the model inconsistently uses to differentiate cases of heart disease versus non-heart disease. It shows how the model recognizes the importance of characteristics at the population level while also examining the reasoning for individual-level importance, directing clinicians and researchers toward consistent priority risk factors. It also aids clinical feasibility of validating the model by showing machine-driven importance in the context of available medical knowledge.

The plot in the Fig. 7 ranks all input features according to their mean absolute SHAP values, giving a global ranking of model feature importance [48]. The higher it ranks a particular feature, the stronger the model statistically supports that feature to influence model predictions. In the figure displayed, features such as 'cp', 'oldpeak', 'thal' and 'ca' are typically higher ranked, which demonstrates their importance in evaluating which patients are at increased risk of cardiovascular events. This global ranking gives a summary of which features are important from a model point of view; additionally, it

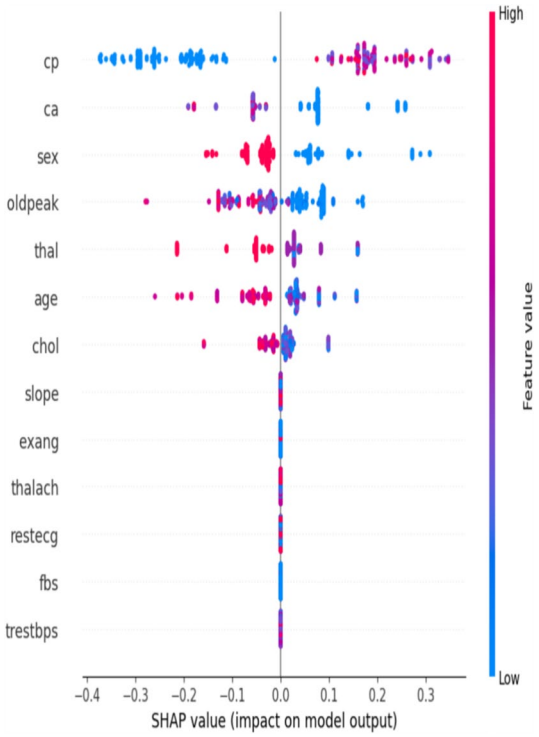


Fig. 8 Summary plot of the proposed ensemble approach

contributes to determining which clinical variables should be emphasized when assessing a patient. In practical terms, the figure provides guidance to health care professionals to identify in which features they should use their clinical discriminant skills; this process additionally draws attention to features that are not usually utilized as valuable predictors, indicating that this may exist for further exploration. The Fig. 7 represents part of the goals of the XAI process, which refers to user interpretability by providing an abstraction of the behavior of complex models in a simple and easily followed manner.

The type of SHAP summary plot shown in the Fig. 8 offers a more nuanced view by grouping and comparing patients with similar explanation profiles [49]. While a standard summary plot is typically focused on what is of the highest importance for contributions to the output, a summary plot could also involve clustering or stratification by prediction class, where the prediction class could simply be heart disease or no heart disease, capturing patterns of similarities in various subgroups. For example, it might show consistently clustering the patients with high "thal," "cp," and "oldpeak" as high risk, and reveal if they (or their combinations) show up with predictions in diverse patient populations. This layered approach also offers transparency into not only what the model is doing but also how it is combining to generalize its reasoning to diverse patient populations. From a clinical decision support perspective, the ability to demonstrably support unique levels of risk stratification by patients changes how patients with particular subtypes are managed or participate in receiving therapies. And, given that increased decision proximal transparency supports the actionability of AI for risk stratification is a requirement for real-world deployment in healthcare settings, this arguably advances the transparency and actionability of trust in these AI systems. Multiple examples have been used to provide consistency in the results when utilizing the SHAP and LIME methodologies for interpreting how important variables (e.g. cp, oldpeak, thal and

ca) were in establishing a person's elevated risk of a cardiovascular event. This indicates that our interpretation findings hold true across the entire dataset.

4.8.3 Computaional cost

The computational resources required to generate explanations using SHAP and LIME were measured. For TreeSHAP, approximately 0.42 s were required to generate an explanation for each instance (Intel i7, 16 GB RAM), whereas LIME required around 0.18 s under the same environment. Although SHAP is slightly more computationally expensive due to its need to sample multiple combinations of the feature space, both SHAP and LIME remain adequate for small- to medium-sized clinical datasets. For large-scale clinical applications, model-specific SHAP variants (e.g., GPUTreeSHAP) can substantially reduce explanation time. Additionally, the prediction time for each patient using the ensemble approach was under 3 milliseconds, supporting the scalability of these XAI methods for real-time clinical decision-support systems.

5 Conclusion

This paper proposes the efficacy of integrating intelligent ensemble learning with XAI methodologies to improve both the accuracy of predictions and the interpretability in diagnosing heart disease. The proposed soft voting classifier achieved a significant enhancement in predictive performance, recording an accuracy of 89.10%, precision of 83.90%, recall of 95.20% and F1-score of 89.90%, surpassing the performance of individual baseline models. The utilisation of SHAP and LIME offered clear insights into the importance of features such as age, cholesterol levels, and resting blood pressure, thereby fostering greater clinical trust and supporting decision-making processes.

Apart from the previously mentioned improvements, the introduction of SHAP and LIME for generating explanations gave local and global interpretability with feature contribution that are aligned to how clinicians think about features. Features such as cp, oldpeak, thal and ca were the primary features identified as having the largest impact on predictions using SHAP, while patient-specific attributes were included in explanations to allow clinicians to better understand how they reached particular predictions. This level of transparency allows clinicians to validate their reasonings through the use of a model along with identifying risk factors that would help with decision making.

While the study has many major strengths, there are some limitations as well. As discussed in section 4.4.3, computational overhead of SHAP based explanation remains a practical limitation for real-time or large-scale deployment. The ensemble model's average prediction time was less than three milliseconds and, therefore, practical for moderate clinical size practices.

Although explanation methods like SHAP and LIME increase the transparency of predictive models, they need to be implemented in such a way that protects patient confidentiality, deters the inappropriate use of sensitive health data, and avoids the potential unintentional amplification of biases inherent to the datasets involved. Given that the model will be trained using publicly available data sources, it is necessary to identify any potential biases in terms of representativeness and fairness when expanding the utility of the system to various types of patients within clinical settings. Using secure data storage methods, ensuring compliance with applicable healthcare legislation, and implementing responsible artificial intelligence (AI) protocols throughout the development of the

knowledge base for implementation of the predictive models will continue to play an important role in transitioning the technology into a real-world clinical application.

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Data availability
The data used in our study is publicly available on Kaggle and can be accessed through the following link: https://www.kaggle.com/datasets/johnsmith88/heart-disease-dataset

Declarations

Ethics approval and consent to participate
This article does not contain studies with human participants or animals performed by any of the authors. The data set used in this study is publicly available on Kaggle and therefore no institutional ethics approval was required. Not applicable. This study did not involve human participants and all data used were anonymized and publicly accessible.

Consent for publication
Not applicable. The manuscript does not contain individual person's data in any form (including images, videos, or personal details).

Competing interests
The authors declare no Conflict of interest.

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References

1. World Health Organization. Cardiovascular diseases. https://www.who.int/health-topics/cardiovascular-diseases#tab=tab_1.
2. Jan B, Dar MI, Choudhary B, Basist P, Khan R, Alhalmi A. Cardiovascular diseases among Indian older adults: a comprehensive review. Cardiovasc Ther. 2024;1:6894693.
3. Centers for Disease Control and Prevention (CDC). Heart Disease Facts. U.S. Department of Health & Human Services, 2023.
4. Hossain MS, Rahman M, Eva AA, Kabir MM, Shin J. Automatic navigation and self-driving technology in agricultural machinery: a state-of-the-art systematic review. IEEE Access. 2025;13:94370–401. https://doi.org/10.1109/ACCESS.2025.3573324.
5. Moreno-Sánchez PA. Improvement of a prediction model for heart failure survival through explainable artificial intelligence. Front Cardiovasc Med. 2023;10:1219586.
6. Das S, et al. XAI-reduct: accuracy preservation despite dimensionality reduction for heart disease classification using explainable AI. J Supercomput. 2023;79(16):18167–97.
7. Dharmarathne G, Bogahawaththa M, Rathnayake U, Meddage DPP. Integrating explainable machine learning and user-centric model for diagnosing cardiovascular disease: A novel approach. Intell Syst Appl. 2024;23:200428.
8. Hossain M, Rahman M, Ahmed M, Kabir M, Shin J. DeepLabv3Att: Integrating attention mechanisms in DeepLabv3 for enhanced road segmentation. In Proc 3ICT, pp. 711–718, 2024. https://doi.org/10.1109/3ict64318.2024.10824285.
9. Wani NA, Kumar R, Bedi J. DeepXplainer: An interpretable deep learning based approach for lung cancer detection using explainable artificial intelligence. Comput Methods Programs Biomed. 2023;243(June):2024. https://doi.org/10.1016/j.cmpb.2023.107879.
10. Chem-mal TS. Prediction of heart disease using hybrid of CNN-LSTM algorithm. J Surv Fish Sci. 2023;10:5700–10.
11. Rahman M, Hossain MS, Eva AA, Kabir MM, Mridha MF, Shin J. Alzheimer's disease classification with a hybrid CNN-SVM approach on enhanced MRI data. In Proc 3ICT, 2024;50–57. https://doi.org/10.1109/3ict64318.2024.10824633.
12. Paudel P, Karna SK, Saud R, Regmi L, Thapa TB, Bhandari M. Unveiling key predictors for early heart attack detection using machine learning and explainable AI technique with LIME. In: ACM Int Conf Proceeding Ser, 2023;71–78. https://doi.org/10.1145/3629188.3629193.
13. Mamun M, Chowdhury SH, Hossain MM, Khatun MR, Iqbal S. Explainability enhanced liver disease diagnosis technique using tree selection and stacking ensemble-based random forest model. Inf Health. 2025;2:17–40. https://doi.org/10.1016/j.infoh.2025.01.001.
14. Anand A, Kadian T, Shetty MK, Gupta A. Explainable AI decision model for ECG data of cardiac disorders. Biomed Signal Process Control. 2022;75:103584. https://doi.org/10.1016/j.bspc.2022.103584.
15. Hassan MR, et al. Prostate cancer classification from ultrasound and MRI images using deep learning based Explainable Artificial Intelligence. Futur Gener Comput Syst. 2022;127:462–72. https://doi.org/10.1016/j.future.2021.09.030.
16. Chowdhury SH, Mamun M, Shaikat TA, Hussain MI, Iqbal S, Hossain MM. An ensemble approach for artificial neural network-based liver disease identification from optimal features through hybrid modeling integrated with advanced explainable AI. Medinformatics. 2025;2(2):107–19. https://doi.org/10.47852/bonviewMEDIN52024744.

17. Hossain M, Rahman M, Rabbani G, Maua J, Rahman A. Plant leaf disease detection using deep stacking: integrating cnns and gradient boosting for enhanced classification accuracy. Intell Syst Des Appl. 2025. https://doi.org/10.1007/978-981-96-4520-6_2.
18. Raschka S, Patterson J, Nolet C. Machine learning in python: main developments and technology trends in data science, machine learning, and artificial intelligence. Information. 2020;11(4):193. https://doi.org/10.3390/info11040193.
19. El-Sappagh S, et al. Alzheimer's disease progression detection model based on an early fusion of cost-effective multimodal data. Futur Gener Comput Syst. 2021;115:680–99. https://doi.org/10.1016/j.future.2020.10.005.
20. Nafisah SI, Muhammad G. Tuberculosis detection in chest radiograph using convolutional neural network architecture and explainable artificial intelligence. Neural Comput Appl. 2024;36(1):111–31. https://doi.org/10.1007/s00521-022-07258-6.
21. Magesh PR, Myloth RD, Tom RJ. An explainable machine learning model for early detection of Parkinson's disease using LIME on DaTSCAN imagery. Comput Biol Med. 2020;126:104041. https://doi.org/10.1016/j.compbiomed.2020.104041.
22. Chowdhury SH, Hussain MI, Alam Chowdhury MS, Sowket Ali M, Hossain MM, Mamun M. Hepatitis C detection from blood donor data using hybrid deep feature synthesis and interpretable machine learning. In Proc 2nd Int Conf Next-Generation Computing, IoT and Machine Learning (NCIM), Gazipur, Bangladesh, 2025;1–6. https://doi.org/10.1109/NCIM65934.2025.11160156.
23. Yagin FH, et al. Explainable artificial intelligence model for identifying COVID-19 gene biomarkers. Comput Biol Med. 2022;154(November):2023. https://doi.org/10.1016/j.compbiomed.2023.106619.
24. Seethi VDR, et al. An explainable AI approach for diagnosis of COVID-19 using MALDI-ToF mass spectrometry. Expert Syst Appl. 2023;236(August):2024. https://doi.org/10.1016/j.eswa.2023.121226.
25. Alfalahi H, Dias SB, Khandoker AH, Chaudhuri KR, Hadjileontiadis LJ. A scoping review of neurodegenerative manifestations in explainable digital phenotyping. npj Park Dis. 2023;9(1):1–22. https://doi.org/10.1038/s41531-023-00494-0.
26. Alehegn M, Joshi RR, Mulay P. Diabetes analysis and prediction using random forest, KNN, Naïve Bayes, and J48: an ensemble approach. Int J Sci Technol Res. 2019;8(9):1346–54.
27. Jain AK, Tiwari RG, Ujjwal N, Singh A. An explainable machine learning model for lumpy skin disease occurrence detection. In 2022 International conference on data analytics for business and industry (ICDABI), 2022;503–508. IEEE.
28. Das K et al. Optimized feature-driven dengue diagnosis using explainable machine learning approaches. In 2025 International conference on quantum photonics, artificial intelligence, and networking (QPAIN). IEEE, 2025.
29. Mamun M et al. An explainable ensemble learning framework with feature optimization for accurate maternal health risk prediction. In 2025 international conference on quantum photonics, artificial intelligence, and networking (QPAIN). IEEE, 2025.
30. John S. Heart Disease Dataset. Kaggle. https://www.kaggle.com/datasets/johnsmith88/heart-disease-dataset.
31. Cortes C, Vapnik V. Support-vector networks. Mach Learn. 1995;20(3):273–97.
32. Quinlan JR. Induction of decision trees. Mach Learn. 1986;1(1):81–106.
33. Nowozin S. Improved information gain estimates for decision tree induction 2012.
34. Breiman L. Random forests. Mach Learn. 2001;45(1):5–32.
35. Strobl C, Boulesteix AL, Augustin T. Unbiased split selection for classification trees based on the Gini index. Comput Stat Data Anal. 2007;52(1):483–501.
36. Dietterich TG. An experimental comparison of three methods for constructing ensembles of decision trees: bagging, boosting, and randomization. Mach Learn. 2000;40(2):139–57.
37. Fernández-Delgado M, Bagnall A, et al. A probabilistic classifier ensemble weighting scheme based on cross-validated accuracy estimates (CAWPE). Data Min Knowl Disc. 2019;33(6):1674–709.
38. Georgiou HV, Mavroforakis ME. A game-theoretic framework for classifier ensembles using weighted majority voting with local accuracy estimates; 2013.
39. Ribeiro MT, Sameer S, Carlos G. Why should i trust you?" Explaining the predictions of any classifier. In Proceedings of the 22nd ACM SIGKDD international conference on knowledge discovery and data mining, 2016;1135–1144.
40. Schwalbe G, Finzel B. A comprehensive taxonomy for explainable artificial intelligence: a systematic survey of surveys on methods and concepts. Data Min Knowl Disc. 2020;38(5):3043–101.
41. Majhi B, Kashyap A. Explainable AI-driven machine learning for heart disease detection using ECG signal. Appl Soft Comput. 2024;167:112225.
42. Lundberg SM, Lee S-I. A unified approach to interpreting model predictions. Advances in neural information processing systems, 2017;30.
43. Ribeiro MT, Sameer S, Carlos G. Anchors: High-precision model-agnostic explanations. In Proceedings of the AAAI conference on artificial intelligence, 2018;32(1).
44. Lamba K, Rani S. Explainable artificial intelligence for deep learning models in diagnosing brain tumor disorder. In International conference on micro-electronics and telecommunication engineering 2023;149–159. Singapore: Springer Nature Singapore.
45. Aas K, Jullum M, Løland A. Explaining individual predictions when features are dependent More accurate approximations to Shapley values. Artif Intell. 2019;298:103502.
46. Okada Y, Ning Y, Ong MEH. Explainable artificial intelligence in emergency medicine: an overview. Clin Exp Emerg Med. 2023;10(4):354–62.
47. Lundberg SM, Lee S-I. A unified approach to interpreting model predictions. In Proceedings of the 31st international conference on neural information processing systems (NeurIPS), 2017.
48. Casalicchio G, Molnar C, Bischl B. Visualizing the feature importance for black box models. In Joint European conference on machine learning and knowledge discovery in databases, pp. 655–670, 2018.
49. Zhao W, Joshi T, Nair VN, Sudjianto A. Shap values for explaining cnn-based text classification models, 2020.

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